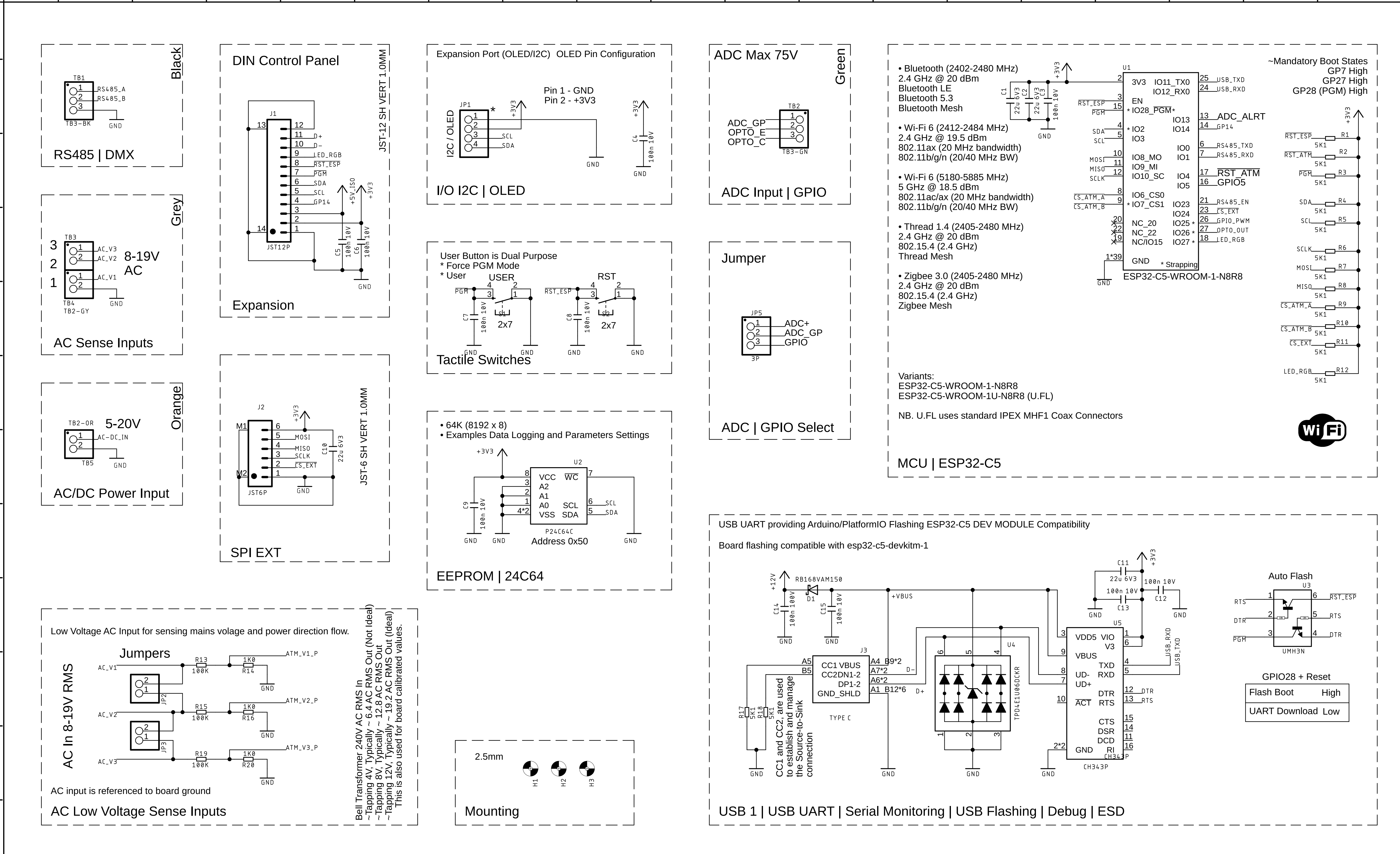

IPEM SIX is an IoT Power Energy Monitor.
IPEM SIX is an ESP32-C5 variant of the IPEM and EPEM series.

For further information and code examples, please see our code and GitHub WIKI pages. ditronix.net | GitHub.com/DitroniX | Hackster.io/DitroniX

Components optimised for BOM

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Production

Enclosure VG-GR34 90x71x57
Main PCB 86 x 66.8

STEM Ambassadors Volunteer

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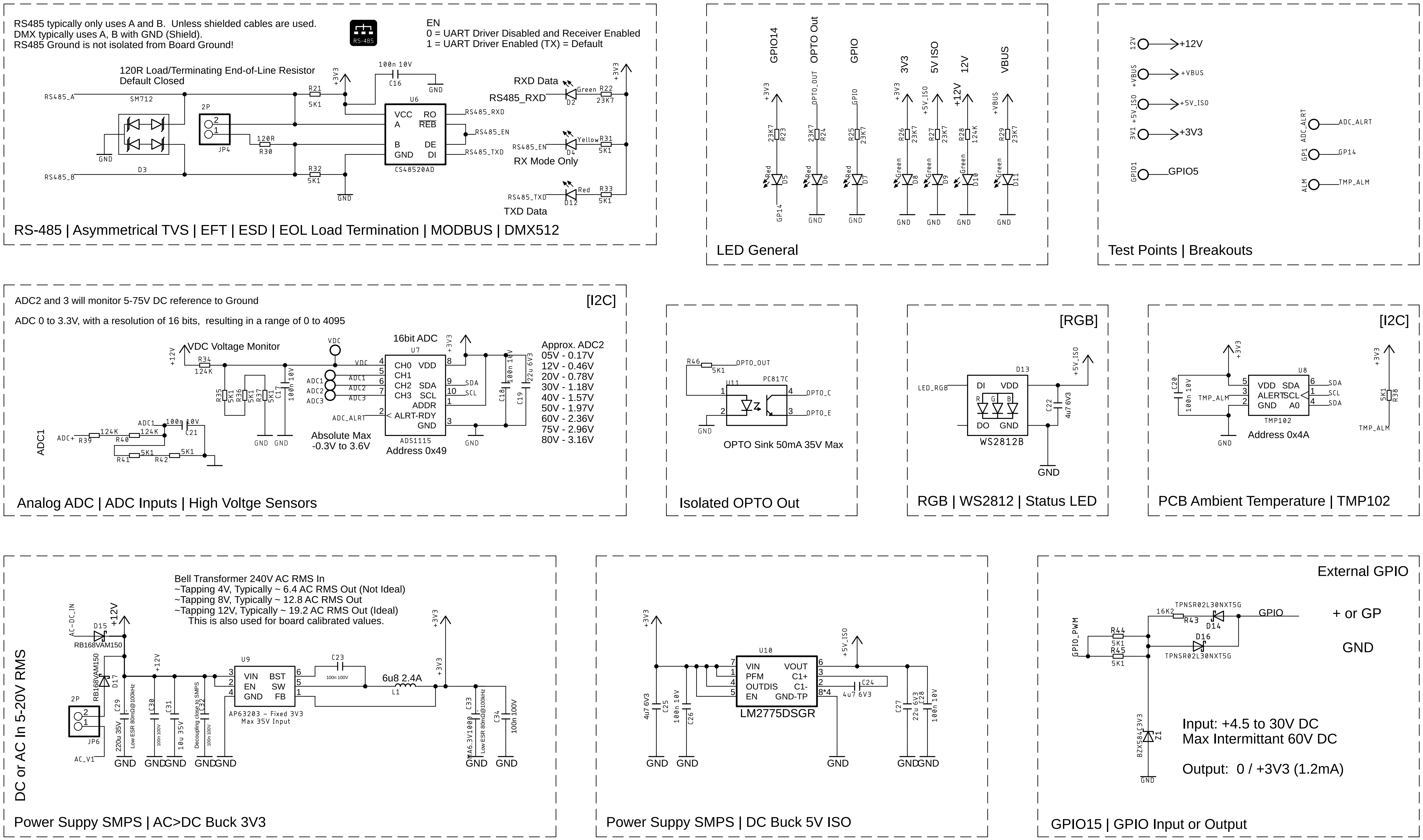
STEM

ESP32-C5 ATM90E32AS SDK ditronix.net DitroniX Ltd - UK	IPEM SIX ESP32-C5-WROOM-1-N8R8 ATM90E32AS 1.2607.102 DUAL-ATM90E32-ESP32-C5-IOT-POWER-ENERGY-MONITOR-BOARD-(PG1) Sheet: 1/4 Drwg/Chk: Dave Williams 09/07/2026 03:31



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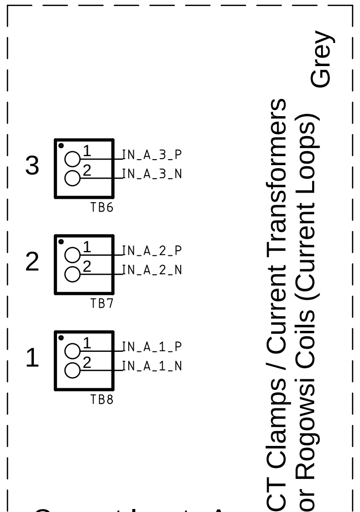
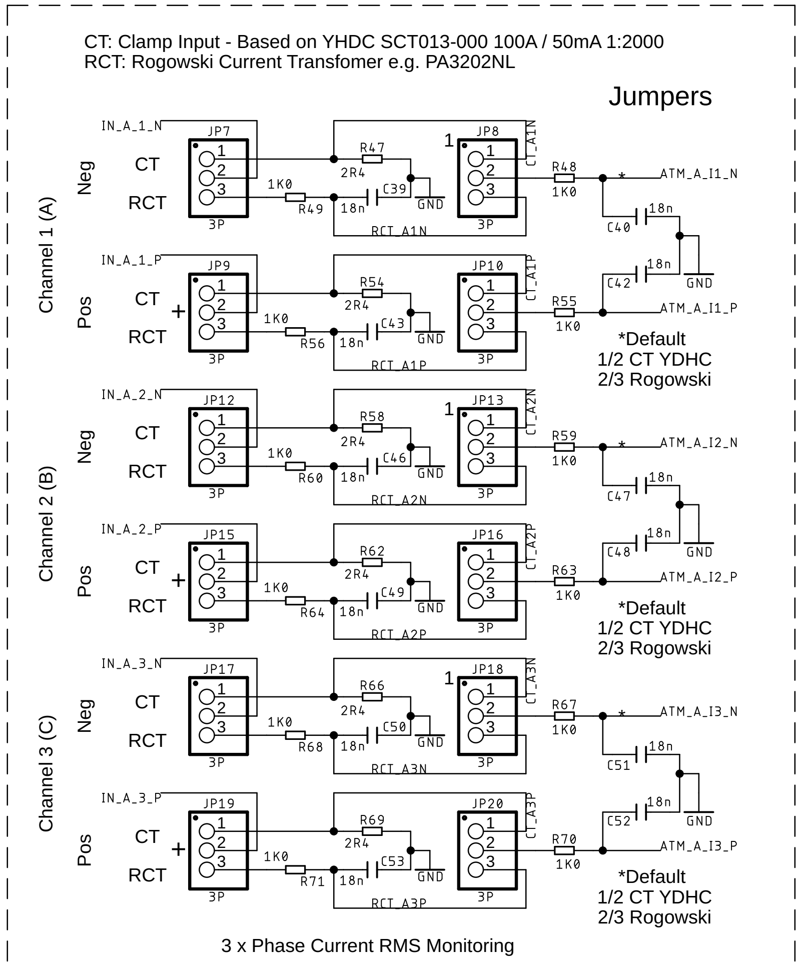
S T E M

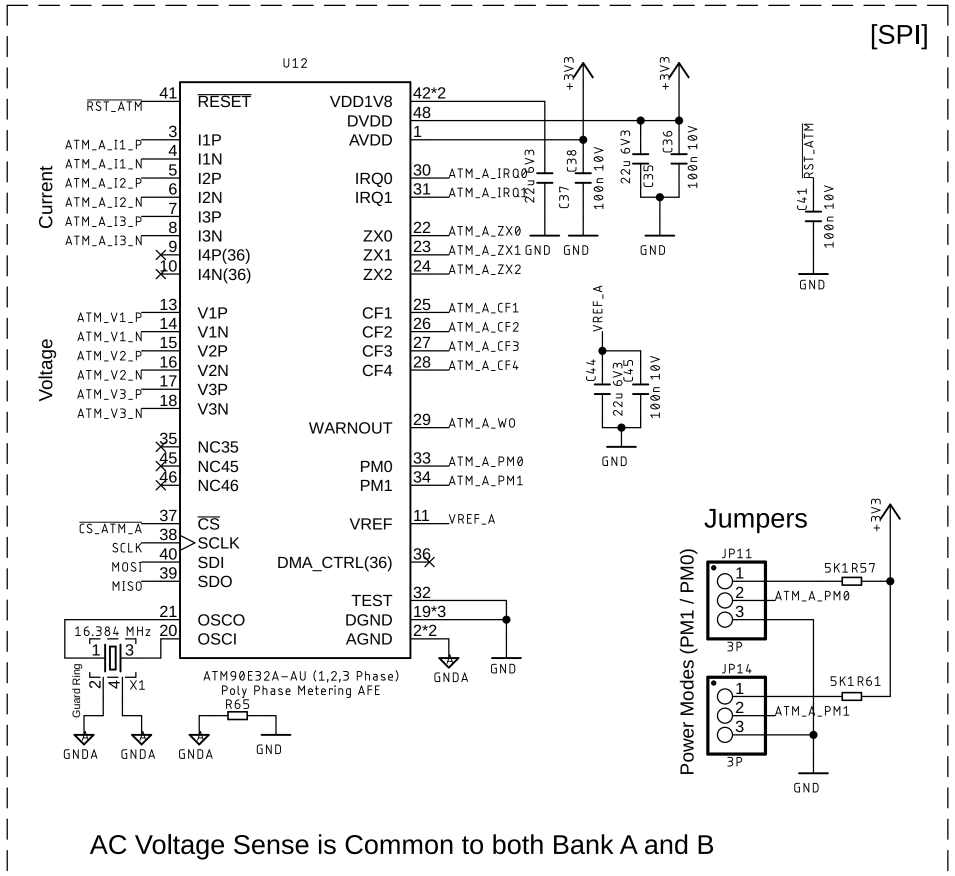
ESP32-C5 ATM90E32AS SDK	IPEM SIX ESP32-C5-WROOM-1-N8R8 ATM90E32AS 1.2607.102		
ditronix.net	DUAL-ATM90E32-ESP32-C5-IOT-POWER-ENERGY-MONITOR-BOARD-(PG2)		
DitroniX Ltd - UK	Sheet: 2/4	Drwg/Chk: Dave Williams	09/07/2026 03:31



Current Inputs A



Current Sensor Inputs Select | CT Clamp | Rogowski



Power Modes (PM1/PM0)

Normal (N) *	1+0 ON
Partial Measure	Only 0 ON
Detection (D)	Only 1 ON
Idle (I)	ALL OFF

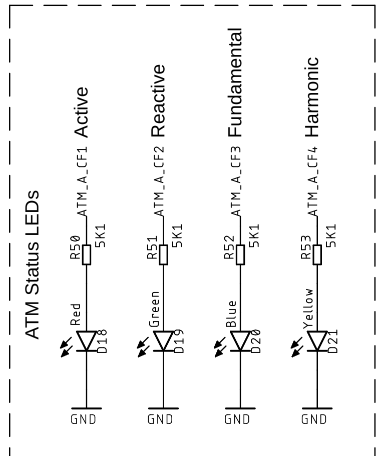
ATM_WO - Warning Out
High = Error

CF1 - Active (Red)
CF2 - ReActive (Green)
CF3 - Fundamental (Blue)
CF4 - Harmonic (Amber/Ylw)
Pulse Outputs Status LEDs

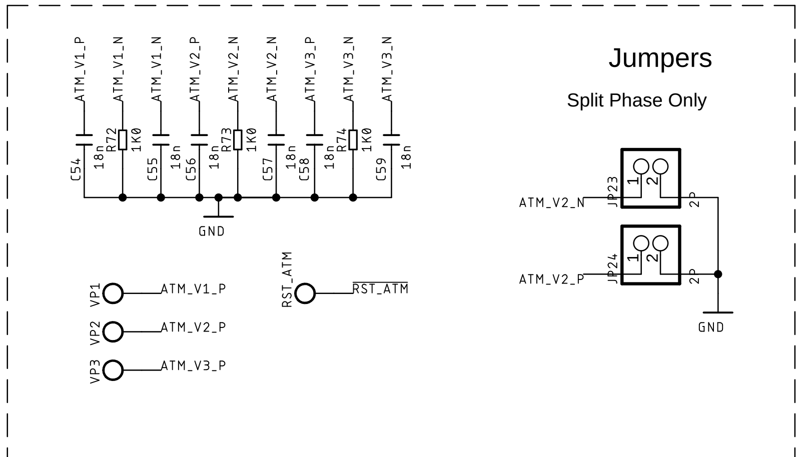
Defaults

WO - L
ZXx - HL
IRQx - L
CFx - L

Mains AC Power Energy Monitor | ATM90E32AS Bank A



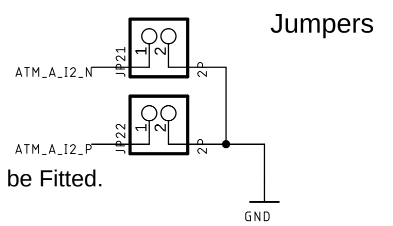
LED ATM90E36 Status



Common to Bank A & B | Split Phase - Voltage (V2)

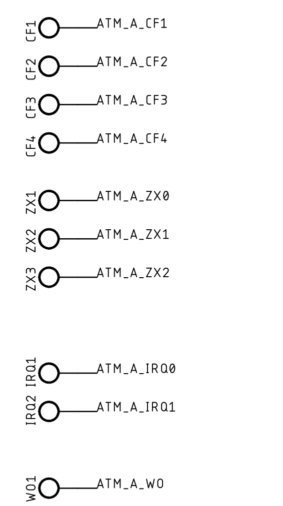
SPLIT Phase ONLY = ON (e.g. USA)

Only Inputs 1 & 3 are used for SPLIT Phase



If Split Phase ALL Current and Voltage Jumpers to be Fitted.

Split Phase - Current (I2)



Test Points

Bank A

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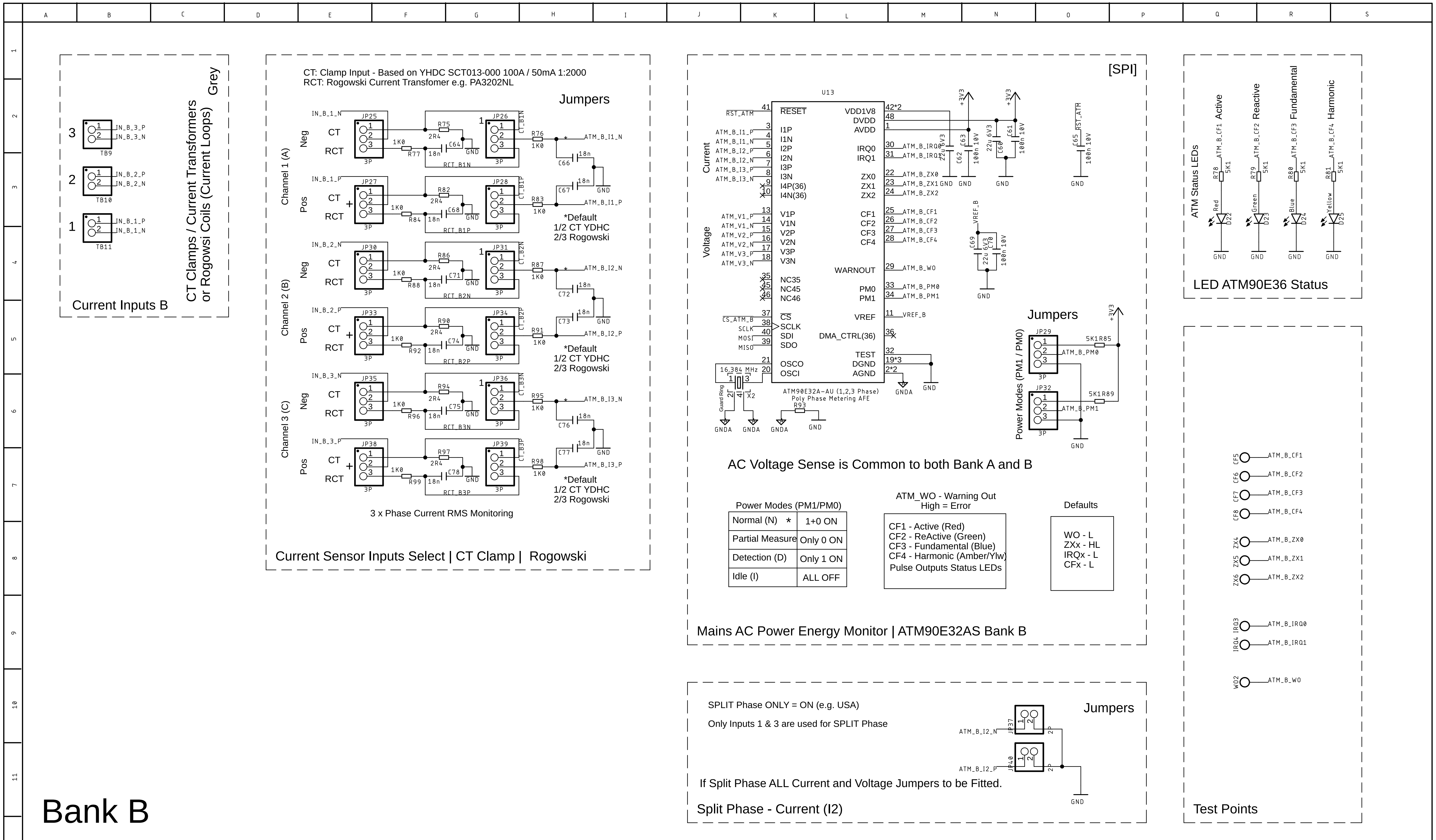
	DUAL-ATM90E32-ESP32-C5-IOT-POWER-ENERGY-MONITOR-BOARD-(PG3)
	Sheet: 3/4 Drwg/Chk: Dave Williams 09/07/2026 03:31



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DUAL-ATM90E32-ESP32-C5-IOT-POWER-ENERGY-MONITOR-BOARD-(PG4)

Sheet:	4/4	Drwg/Chk:	Dave Williams	09/07/2026 03:31